

Implement a Sobel algorithm using Open CV to filter the input image.

Code:

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import files
# Upload image
uploaded = files.upload()
# Read image
img = cv2.imread(list(uploaded.keys())[0], 0)
# Apply Sobel filter
sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=5)
sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=5)
# Display results
print("Original Image")
cv2_imshow(img)
print("Sobel X")
cv2_imshow(np.uint8(np.absolute(sobelx)))
print("Sobel Y")
cv2_imshow(np.uint8(np.absolute(sobely)))
```

Output:



